

Further Pure Mathematics 1

Complete Notes & Exercises

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Contents overview: This document covers all seven chapters of the Further Pure Mathematics 1 syllabus in the order they appear in the textbook. Each chapter contains full derivations with step-by-step algebra, worked examples, TikZ diagrams, and a graded exercise set (Basic / Medium / Hard) followed by two Exploration questions going beyond the standard syllabus.

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Chapter 1

Roots of Polynomial Equations

The roots of a polynomial are intimately related to its coefficients via symmetric functions. This chapter derives those relations for quadratics, cubics, and quartics, then applies substitution techniques to find new equations with transformed roots.

1 Quadratics

1.1 Vieta's Formulae for a Quadratic

Consider the monic quadratic with roots α, β :

$$(x - \alpha)(x - \beta) = x^2 - (\alpha + \beta)x + \alpha\beta.$$

Comparing with the general form $ax^2 + bx + c = 0$ (divide through by a):

Result 1.1 — Symmetric functions of quadratic roots

If α, β are the roots of $ax^2 + bx + c = 0$, then

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

Derivation

Expanding $(x - \alpha)(x - \beta) = 0$ and matching coefficients with $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$:

$$\begin{aligned} x^2 - (\alpha + \beta)x + \alpha\beta &= x^2 + \frac{b}{a}x + \frac{c}{a} \\ \Rightarrow \quad \alpha + \beta &= -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}. \quad \square \end{aligned}$$

1.2 Higher Symmetric Functions

From the two elementary relations above, higher combinations can be derived without finding α, β explicitly.

Result 1.2

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta, \quad \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta).$$

Derivations

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta.$$

$$\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2) = (\alpha + \beta)[(\alpha + \beta)^2 - 3\alpha\beta]. \quad \square$$

Example 1.1 — Given roots of $2x^2 - 5x + 1 = 0$, find $\alpha^2 + \beta^2$

$$\alpha + \beta = \frac{5}{2}, \quad \alpha\beta = \frac{1}{2}.$$

$$\alpha^2 + \beta^2 = \left(\frac{5}{2}\right)^2 - 2 \cdot \frac{1}{2} = \frac{25}{4} - 1 = \frac{21}{4}.$$

2 Cubics

2.1 Vieta's Formulae for a Cubic

Expanding $(x - \alpha)(x - \beta)(x - \gamma)$:

$$x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma.$$

Result 2.1 — Symmetric functions of cubic roots

If α, β, γ are the roots of $ax^3 + bx^2 + cx + d = 0$, then

$$\alpha + \beta + \gamma = -\frac{b}{a}, \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}, \quad \alpha\beta\gamma = -\frac{d}{a}.$$

Key derived identities

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha).$$

$$\alpha^3 + \beta^3 + \gamma^3 = (\alpha + \beta + \gamma)^3 - 3(\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \gamma\alpha) + 3\alpha\beta\gamma.$$

Example 2.1 — Roots of $x^3 - 6x^2 + 11x - 6 = 0$

Here $a = 1, b = -6, c = 11, d = -6$, so

$$\alpha + \beta + \gamma = 6, \quad \alpha\beta + \beta\gamma + \gamma\alpha = 11, \quad \alpha\beta\gamma = 6.$$

$$\alpha^2 + \beta^2 + \gamma^2 = 36 - 22 = 14.$$

(Roots are 1, 2, 3; check: $1 + 4 + 9 = 14$. ✓)

3 Quartics

3.1 Vieta's Formulae for a Quartic

Expanding $\prod_{i=1}^4 (x - \alpha_i)$ and comparing with $ax^4 + bx^3 + cx^2 + dx + e = 0$:

Result 3.1

$$\sum \alpha_i = -\frac{b}{a}, \quad \sum_{i < j} \alpha_i \alpha_j = \frac{c}{a}, \quad \sum_{i < j < k} \alpha_i \alpha_j \alpha_k = -\frac{d}{a}, \quad \alpha_1 \alpha_2 \alpha_3 \alpha_4 = \frac{e}{a}.$$

Example 3.1 — Find $\sum \alpha^2$ for roots of $x^4 - 5x^3 + 3x^2 + x - 2 = 0$

$$\sum \alpha = 5, \quad \sum_{i < j} \alpha_i \alpha_j = 3.$$

$$\sum \alpha^2 = \left(\sum \alpha \right)^2 - 2 \sum_{i < j} \alpha_i \alpha_j = 25 - 6 = 19.$$

4 Substitutions

To find a new equation whose roots are $f(\alpha), f(\beta), \dots$ (a function of the original roots), substitute $x = f^{-1}(y)$ into the original equation, simplify, and relabel $y \rightarrow x$.

Example 4.1 — Roots of $x^3 - 3x + 1 = 0$ are α, β, γ . Find the equation with roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1$.

Let $y = 2x + 1$, so $x = \frac{y-1}{2}$. Substitute:

$$\left(\frac{y-1}{2} \right)^3 - 3 \cdot \frac{y-1}{2} + 1 = 0$$

$$\frac{(y-1)^3}{8} - \frac{3(y-1)}{2} + 1 = 0.$$

Multiply by 8:

$$(y-1)^3 - 12(y-1) + 8 = 0.$$

Expand $(y-1)^3 = y^3 - 3y^2 + 3y - 1$:

$$y^3 - 3y^2 + 3y - 1 - 12y + 12 + 8 = 0 \implies y^3 - 3y^2 - 9y + 19 = 0.$$

Replace y by x : the required equation is $x^3 - 3x^2 - 9x + 19 = 0$. $\square$

Example 4.2 — Roots $\frac{1}{\alpha^2}$ from $2x^3 - x^2 + 3x - 5 = 0$

Let $y = \frac{1}{x^2}$, so $x = y^{-1/2}$. Equivalently, substitute $x^2 = \frac{1}{y}$ and use $x = \pm \frac{1}{\sqrt{y}}$, then square to eliminate the surd:

Step 1 — isolate odd-power terms:

$$2x^3 + 3x = x^2 + 5 \implies x(2x^2 + 3) = x^2 + 5.$$

Step 2 — square both sides:

$$x^2(2x^2 + 3)^2 = (x^2 + 5)^2.$$

Step 3 — substitute $x^2 = \frac{1}{y}$:

$$\frac{1}{y} \left(\frac{2}{y} + 3 \right)^2 = \left(\frac{1}{y} + 5 \right)^2.$$

Multiply through by y^3 :

$$(2 + 3y)^2 = (1 + 5y)^2 y \implies 9y^3 + 5y^2 - 8y - 4 = 0$$

(after expansion and simplification).

Exercise — Chapter 1 — Roots of Polynomial Equations

Basic

Q1. The equation $3x^2 - 7x + 2 = 0$ has roots α, β . Without solving, find: (a) $\alpha + \beta$; (b) $\alpha\beta$; (c) $\alpha^2 + \beta^2$; (d) $\frac{1}{\alpha} + \frac{1}{\beta}$. [5 marks]

Q2. The cubic $x^3 - 6x^2 + 11x - 6 = 0$ has roots α, β, γ . Find: (a) $\alpha + \beta + \gamma$; (b) $\alpha\beta + \beta\gamma + \gamma\alpha$; (c) $\alpha\beta\gamma$; (d) $\alpha^2 + \beta^2 + \gamma^2$. [5 marks]

Q3. The equation $x^2 - 5x + 3 = 0$ has roots α, β . Find the quadratic equation with roots $3\alpha - 1$ and $3\beta - 1$, giving your answer in the form $x^2 + px + q = 0$. [4 marks]

Q4. Write down Vieta's formulae for a quartic $x^4 + px^3 + qx^2 + rx + s = 0$ with roots $\alpha, \beta, \gamma, \delta$. Hence show that $\sum \alpha^2 = p^2 - 2q$. [4 marks]

Q5. The roots of $x^3 + 2x - 9 = 0$ are α, β, γ . Show that $\alpha^3 + \beta^3 + \gamma^3 = 27$. [4 marks]

Medium

Q6. The equation $2x^3 - 3x^2 + x - 5 = 0$ has roots α, β, γ . Find the cubic equation whose roots are $\alpha + 1, \beta + 1, \gamma + 1$. [6 marks]

Q7. Given that the equation $x^4 - 2x^3 - x + k = 0$ has two equal roots, find all possible values of k . [7 marks]

Q8. The roots of $x^3 - 4x + 1 = 0$ are α, β, γ .

(a) Find $\alpha^2 + \beta^2 + \gamma^2$. [2]

(b) Show that $\alpha^4 + \beta^4 + \gamma^4 = 24$. [4]

(c) Find the cubic equation with roots $\alpha^2, \beta^2, \gamma^2$. [4]

[10 marks]

Q9. The quartic $x^4 - 3x^3 + ax^2 + bx + 4 = 0$ has a root $x = 2$ of multiplicity 2. Find a and b , and hence find all four roots. [8 marks]

Q10. For the equation $x^3 + px + q = 0$ with roots α, β, γ , prove that $\alpha^3 + \beta^3 + \gamma^3 = -3q$. [5 marks]

Hard

Q11. The equation $x^4 - 10x^2 + 1 = 0$ has roots $\alpha, \beta, \gamma, \delta$.

(a) Find all four elementary symmetric polynomials. [3]

(b) Show that $\alpha^4 + \beta^4 + \gamma^4 + \delta^4 = 194$. [5]

(c) Find the quartic whose roots are $\alpha + \frac{1}{\alpha}, \beta + \frac{1}{\beta}, \gamma + \frac{1}{\gamma}, \delta + \frac{1}{\delta}$. [6]

[14 marks]

Q12. The equation $f(x) = x^3 + ax + b$ has roots α, β, γ .

(a) Prove that $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = a^2$. [3]

(b) Hence, or otherwise, find the cubic equation with roots $\alpha^2\beta^2, \beta^2\gamma^2, \gamma^2\alpha^2$. [5]

(c) Given that one root of the original equation is double another, show that $4a^3 + 27b^2 = 0$. [6]

[14 marks]

Q13. Prove Newton's identity: if $p_k = \alpha^k + \beta^k + \gamma^k$ for roots of $x^3 + px + q = 0$, show that

$$p_k - p p_{k-2} - q p_{k-3} = 0, \quad k \geq 3,$$

and use it to compute p_3, p_4, p_5 in terms of p and q . [10 marks]

Q14. The equation $x^4 + bx^2 + c = 0$ has four real roots forming an arithmetic progression with common difference d . Show that $b = -5d^2$ and $c = \frac{9d^4}{16}$, and find the four roots in terms of d . [9 marks]

Q15. The equations $x^3 + px + q = 0$ and $x^3 + p'x + q' = 0$ have a common root α . By considering α^3 from both equations, show that $\alpha = \frac{q' - q}{p - p'}$ (assuming $p \neq p'$), and hence find a condition on p, q, p', q' for a common root to exist. [9 marks]

Exploration (Beyond Syllabus)

Q16. The discriminant and its meaning.

The *discriminant* of a cubic $x^3 + px + q = 0$ is defined as $\Delta = -4p^3 - 27q^2$.

(a) Using Vieta's formulae, show that

$$\prod_{i < j} (\alpha_i - \alpha_j)^2 = (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$$

can be expressed purely in terms of p and q . [This is Δ .] [6]

(b) Hence explain: (i) why $\Delta > 0$ implies three distinct real roots; (ii) why $\Delta = 0$ implies a repeated root; (iii) what $\Delta < 0$ tells you about the nature of the roots. [3]

(c) Compute Δ for $x^3 - 3x + 2 = 0$. Verify by factorising the cubic directly. [3]

[12 marks]

Q17. Power sums and the recurrence (Newton–Girard).

Define $e_1 = \sum \alpha_i$, $e_2 = \sum_{i < j} \alpha_i \alpha_j$, $e_3 = \alpha \beta \gamma$ (the elementary symmetric polynomials) and $p_k = \sum \alpha_i^k$.

(a) Prove Newton's recurrence:

$$p_k = e_1 p_{k-1} - e_2 p_{k-2} + e_3 p_{k-3}, \quad k \geq 4,$$

by multiplying the relation $\alpha^3 = e_1 \alpha^2 - e_2 \alpha + e_3$ by α^{k-3} , summing over all three roots, and using the base cases. [7]

(b) Use the recurrence to find $p_1, \dots, p_6$ for the roots of $x^3 - x - 1 = 0$. [5]

(c) Show that p_k is always an integer for these roots, even though the roots themselves are irrational. [3]

[15 marks]

Chapter 2

Rational Functions

A *rational function* is a ratio of two polynomials $\frac{P(x)}{Q(x)}$. Their graphs are characterised by asymptotes — lines the graph approaches but never crosses (or crosses only finitely many times).

5 Vertical Asymptotes

Definition 5.1. The line $x = a$ is a vertical asymptote of $y = f(x)$ if $|f(x)| \rightarrow \infty$ as $x \rightarrow a$.

For a rational function $\frac{P(x)}{Q(x)}$ in lowest terms, vertical asymptotes occur at the zeros of $Q(x)$.

Result 5.1

If $Q(a) = 0$ and $P(a) \neq 0$, then $x = a$ is a vertical asymptote of $\frac{P(x)}{Q(x)}$.

Example 5.1 — Vertical asymptotes of $y = \frac{x+1}{(x-2)(x+3)}$

Zeros of denominator: $x = 2$ and $x = -3$. Check numerator: $P(2) = 3 \neq 0$ and $P(-3) = -2 \neq 0$. Vertical asymptotes: $x = 2$ and $x = -3$.

Example 5.2 — $y = \frac{x^2 - 4}{x - 2}$: no vertical asymptote at $x = 2$

$$y = \frac{(x-2)(x+2)}{x-2} = x+2 \quad (x \neq 2).$$

The factor cancels — $x = 2$ is a removable discontinuity (hole), not an asymptote.

6 Oblique Asymptotes

Definition 6.1. The line $y = mx + c$ is an oblique (slant) asymptote of $y = f(x)$ if

$$f(x) - (mx + c) \rightarrow 0 \quad \text{as } x \rightarrow \pm\infty.$$

When $m = 0$ the asymptote is horizontal ($y = c$).

Result 6.1 — Finding oblique asymptotes by polynomial division

If $\deg P = \deg Q + 1$, perform long division:

$$\frac{P(x)}{Q(x)} = (mx + c) + \frac{R(x)}{Q(x)},$$

where $\deg R < \deg Q$. As $x \rightarrow \infty$, $\frac{R(x)}{Q(x)} \rightarrow 0$, so $y = mx + c$ is the oblique asymptote.

If $\deg P < \deg Q$: horizontal asymptote $y = 0$.

If $\deg P = \deg Q$: horizontal asymptote $y =$ ratio of leading coefficients.

Example 6.1 — Oblique asymptote of $y = \frac{x^2 + 3x - 1}{x + 2}$

Divide $x^2 + 3x - 1$ by $x + 2$:

$$x^2 + 3x - 1 = (x + 2)(x + 1) - 3.$$

So $y = (x + 1) - \frac{3}{x + 2}$. As $x \rightarrow \infty$, the remainder $\rightarrow 0$, giving oblique asymptote $y = x + 1$.

Example 6.2 — Horizontal asymptote of $y = \frac{3x^2 - 1}{x^2 + 2x}$

Leading coefficients: numerator 3, denominator 1. Horizontal asymptote: $y = 3$.

7 Inequalities Involving Rational Functions

To solve $\frac{P(x)}{Q(x)} > 0$ (or < 0), the key principle is:

- **Never multiply both sides by $Q(x)$** if $Q(x)$ can be negative — its sign is unknown.
- Instead, multiply by $[Q(x)]^2 > 0$ (always positive), or use a sign diagram.

Result 7.1 — Sign diagram method

1. Find all zeros and poles (vertical asymptotes).
2. Arrange them on a number line.
3. Determine the sign of the expression in each interval.

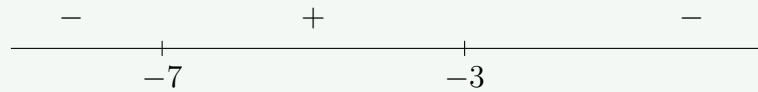
Example 7.1 — Solve $\frac{x - 1}{x + 3} > 2$

Rearrange to have zero on the right:

$$\frac{x - 1}{x + 3} - 2 > 0 \implies \frac{x - 1 - 2(x + 3)}{x + 3} > 0 \implies \frac{-x - 7}{x + 3} > 0.$$

Critical values: $x = -7$ (zero) and $x = -3$ (pole).

Sign diagram for $\frac{-(x+7)}{x+3}$:



The expression is > 0 when $-7 < x < -3$.

8 Relationships Between Curves

Given $y = f(x)$, related curves can be sketched by transformation:

Result 8.1

- $y = |f(x)|$: reflect any portion with $f(x) < 0$ in the x -axis.
- $y = f(|x|)$: keep the right half ($x > 0$) of $y = f(x)$, reflect it in the y -axis.
- $y^2 = f(x)$: defined only where $f(x) \geq 0$; for each point (x_0, y_0) on $y = \sqrt{f(x)}$, also plot $(x_0, -y_0)$.
- $y = \frac{1}{f(x)}$: zeros of f become vertical asymptotes; asymptotes of f become zeros; where f is large, $1/f$ is small; where f is small and positive, $1/f$ is large.

Example 8.1 — Sketch $y = \frac{1}{x^2 - 1}$ from $y = x^2 - 1$

$y = x^2 - 1$ has zeros at $x = \pm 1$ and vertex $(0, -1)$. For $y = \frac{1}{x^2 - 1}$:

- Vertical asymptotes at $x = \pm 1$.
- Horizontal asymptote $y = 0$.
- At $x = 0$: $y = \frac{1}{-1} = -1$.
- For $|x| > 1$: $y > 0$; for $|x| < 1$: $y < 0$.

Exercise — Chapter 2 — Rational Functions

Basic

Q1. State the equations of all asymptotes of each function:

(a) $y = \frac{3}{x-5}$ [2]

(b) $y = \frac{x+2}{x^2-9}$ [3]

$$(c) \ y = \frac{2x^2 + 1}{x^2 - 4} \quad [3]$$

[8 marks]

Q2. Find the oblique asymptote of $y = \frac{x^2 - 2x + 3}{x - 1}$ by polynomial long division. [4 marks]

Q3. Solve the inequality $\frac{x + 4}{x - 1} < 3$. [5 marks]

Q4. Sketch $y = \frac{1}{x^2 - 4}$, labelling all intercepts, asymptotes, and indicating the sign in each region. [5 marks]

Q5. Describe the transformation from $y = f(x)$ to $y = |f(x)|$ and from $y = f(x)$ to $y = f(|x|)$. Illustrate both with $f(x) = x - 2$. [4 marks]

Medium

Q6. The curve C has equation $y = \frac{x^2 - x - 2}{x - 3}$.

(a) Find all asymptotes. [3]

(b) Find the coordinates of any stationary points. [5]

(c) Sketch C . [3]

[11 marks]

Q7. Solve the inequality $\frac{x^2 - 1}{x^2 + x - 6} > 0$ using a sign diagram. [7 marks]

Q8. Given $y = \frac{ax + b}{cx + d}$, find conditions on a, b, c, d such that:

(a) The curve passes through the origin. [1]

(b) The horizontal asymptote is $y = 2$. [1]

(c) The curve has no vertical asymptote (state what happens instead). [3]

(d) The curve is self-inverse: $f(f(x)) = x$. [4]

[9 marks]

Q9. Sketch, on separate axes, the graphs of $y^2 = \frac{x}{x - 1}$, stating the domain clearly. [6 marks]

Q10. Show that the curve $y = \frac{x^2 + kx + 1}{x}$ has no stationary points if and only if $-2 < k < 2$. [6 marks]

Hard

Q11. The curve $y = \frac{x^2 - 2x - 3}{(x - 1)^2}$ has a stationary point.

(a) Find the stationary point(s) and determine their nature. [6]

(b) Find all asymptotes and any points where the curve crosses the horizontal asymp-

tote.

[5]

(c) Sketch the curve.

[3]

[14 marks]

Q12. Prove that the curve $y = \frac{x^2 + ax + 1}{x^2 + bx + 1}$ always lies between two horizontal lines for any real constants a, b with $a \neq b$, and find those lines. [9 marks]

Q13. Solve $\frac{x+1}{x-2} > \frac{x-1}{x+2}$, showing all steps clearly without squaring. [8 marks]

Q14. A curve has equation $y = \frac{2x^2 - 5x + 2}{x^2 - 4}$.

(a) Show that the curve has a removable discontinuity at one value of x and a vertical asymptote at another. [4]

(b) Find the range of y . [6]

[10 marks]

Q15. Prove that for the function $y = \frac{x^2 + px + q}{x + r}$ to have two distinct real stationary points, we need $p^2 - 2pr + r^2 > 4q - 4r^2$. Deduce a simpler condition when $p = r$. [9 marks]

Exploration (Beyond Syllabus)

Q16. Continued fractions and rational approximations.

A *continued fraction* is an expression of the form

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \cdots}}}$$

(a) The rational function $f_n(x) = \frac{x^{n+1} - 1}{x^n - 1}$ can be written as a continued fraction.

Show that $f_2(x) = x + \frac{1}{x + \frac{1}{x}}$ and verify algebraically. [4]

(b) Define the sequence of rational functions $r_n = \frac{F_{n+1}}{F_n}$ where F_n is the n -th Fibonacci number. Show that each r_n is a convergent of the continued fraction for $\phi = \frac{1 + \sqrt{5}}{2}$. [5]

(c) Prove that $|r_n - \phi| < \frac{1}{F_n^2}$. [Hint: use the identity $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$.] [5]

[14 marks]

Q17. Partial fractions and residues.

Partial fraction decomposition is the algebraic counterpart of the complex analysis technique of residues.

- (a) Decompose $\frac{1}{(x-a)(x-b)(x-c)}$ into partial fractions (assume a, b, c distinct).

Express the coefficient of $\frac{1}{x-a}$ in terms of a, b, c . [4]

- (b) The *residue* of $f(x) = \frac{P(x)}{Q(x)}$ at a simple pole $x = a$ is $\lim_{x \rightarrow a} (x-a)f(x)$. Show this equals the partial fraction coefficient of $\frac{1}{x-a}$. [3]

- (c) Use residues to show that $\sum_{k=0}^{n-1} \frac{1}{(k+a)(k+b)} = \frac{1}{b-a} [\psi(n+b) - \psi(b) - \psi(n+a) + \psi(a)]$ where ψ is the digamma function, by interpreting the sum via telescoping after partial fraction decomposition. [6]

Note: you are not expected to know ψ — just show how the partial fraction structure leads naturally to this form.

[13 marks]

Chapter 3

Summation of Series

9 Standard Summation Results

9.1 Sum of the First n Natural Numbers

Result 9.1

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Derivation

Let $S = 1 + 2 + \cdots + n$. Write forwards and backwards then add:

$$S = 1 + 2 + \cdots + (n-1) + n$$

$$S = n + (n-1) + \cdots + 2 + 1$$

$$2S = (n+1) + (n+1) + \cdots + (n+1) = n(n+1).$$

Hence $S = \frac{n(n+1)}{2}$. $\square$

Example 9.1 — Check $n = 4$

$$1 + 2 + 3 + 4 = 10 = \frac{4 \cdot 5}{2} = 10. \checkmark$$

9.2 Sum of Squares

Result 9.2

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Derivation via telescoping

Use $k^3 - (k-1)^3 = 3k^2 - 3k + 1$. Sum from $k = 1$ to n ; the left side telescopes to n^3 :

$$\begin{aligned} n^3 &= 3 \sum_{k=1}^n k^2 - 3 \cdot \frac{n(n+1)}{2} + n \\ 3 \sum_{k=1}^n k^2 &= n^3 + \frac{3n^2 + n}{2} = \frac{n(2n+1)(n+1)}{2} \\ \sum_{k=1}^n k^2 &= \frac{n(n+1)(2n+1)}{6}. \quad \square \end{aligned}$$

9.3 Sum of Cubes

Result 9.3

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Derivation

Use $k^4 - (k-1)^4 = 4k^3 - 6k^2 + 4k - 1$, telescope to get n^4 , substitute known results:

$$\begin{aligned} 4 \sum_{k=1}^n k^3 &= n^4 + n^2(2n+1) = n^2(n+1)^2, \\ \sum_{k=1}^n k^3 &= \frac{n^2(n+1)^2}{4} = \left[\frac{n(n+1)}{2} \right]^2. \quad \square \end{aligned}$$

10 Method of Differences

10.1 Telescoping Principle

If $u_r = f(r) - f(r-1)$, then

$$\sum_{r=1}^n u_r = f(n) - f(0),$$

since all intermediate terms cancel. This is the *method of differences* (telescoping).

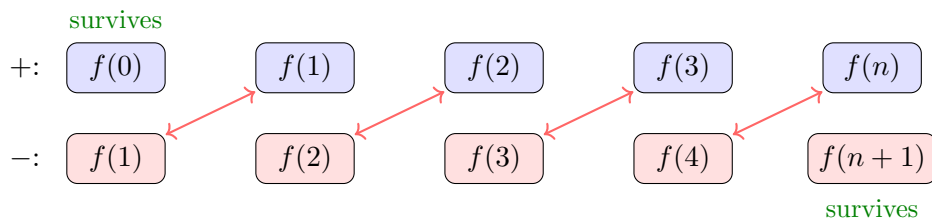


Figure 1. Cancellation pattern: only $+f(0)$ and $-f(n+1)$ survive (or equivalently $f(n) - f(0)$ depending on orientation).

10.2 Worked Examples

Example 10.1 — $\sum_{r=1}^n \frac{1}{r(r+1)}$ via partial fractions

$$\frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{r+1} = f(r-1) - f(r), \quad f(r) = \frac{1}{r+1}.$$

$$\sum_{r=1}^n \frac{1}{r(r+1)} = f(0) - f(n) = 1 - \frac{1}{n+1} = \frac{n}{n+1}.$$

As $n \rightarrow \infty$, sum $\rightarrow 1$.

Example 10.2 — $\sum_{r=1}^n r(r+1)$ via method of differences

Note $r(r+1)(r+2) - (r-1)r(r+1) = 3r(r+1)$. So

$$\sum_{r=1}^n r(r+1) = \frac{1}{3} [n(n+1)(n+2) - 0] = \frac{n(n+1)(n+2)}{3}.$$

11 Converging Series

Definition 11.1. The n -th partial sum is $S_n = \sum_{r=1}^n a_r$. A series converges if $\lim_{n \rightarrow \infty} S_n = L \in \mathbb{R}$; otherwise it diverges.

Result 11.1 — Necessary condition

If $\sum a_r$ converges, then $a_r \rightarrow 0$.

Result 11.2 — Geometric series

$$\sum_{r=0}^{\infty} ar^k = \frac{a}{1-r} \text{ provided } |r| < 1; \text{ diverges if } |r| \geq 1.$$

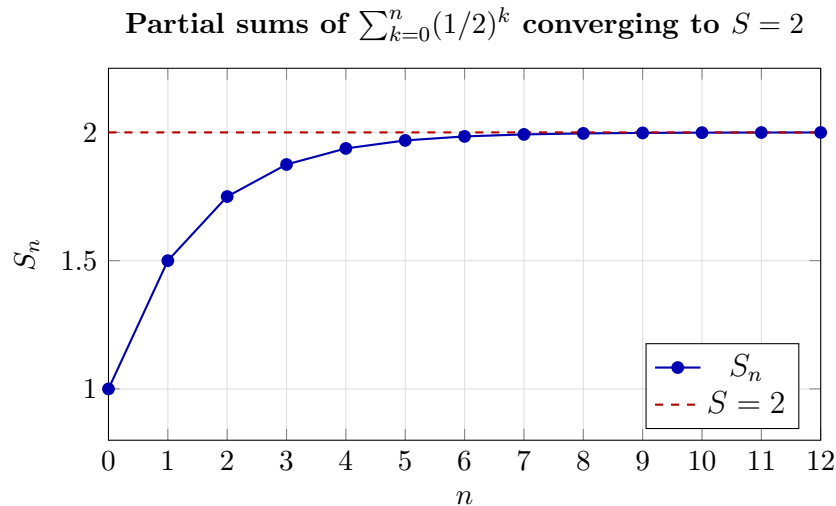


Figure 2. Partial sums of a geometric series approaching the limit $S = 2$.

Exercise — Chapter 3 — Summation of Series

Basic

- Q1.** Evaluate using standard results: (a) $\sum_{r=1}^n r$; (b) $\sum_{r=1}^n r^2$; (c) $\sum_{r=1}^n r^3$; (d) $\sum_{r=1}^{10} r$; (e) $\sum_{r=1}^6 r^2$. [5 marks]
- Q2.** Find closed forms for: (a) $\sum_{r=1}^n (3r-1)$; (b) $\sum_{r=1}^n (2r^2+5)$; (c) $\sum_{r=1}^n (r^2-r+1)$. Factorise fully. [6 marks]
- Q3.** Show that $3 + 6 + 12 + \dots + 3 \cdot 2^{n-1} = 3(2^n - 1)$ and verify for $n = 4$. [4 marks]
- Q4.** A geometric series has $a = 5$, $r = 3/5$. Find S_n , S_∞ , and the least n such that $S_\infty - S_n < 0.01$. [6 marks]
- Q5.** Show by writing out terms for $n = 5$ that $\sum_{r=1}^n [f(r) - f(r-1)] = f(n) - f(0)$. Hence find $\sum_{r=1}^n (2r-1)$ using $f(r) = r^2$. [4 marks]

Medium

- Q6.** Find $\sum_{r=1}^n r(r+1)(r+2)$ in fully factorised form. [6 marks]
- Q7.** Using partial fractions, find $\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)}$ and state its sum to infinity. [7 marks]
- Q8.** (a) Show $f(r) - f(r-1) = 2r$ for $f(r) = r(r+1)$. (b) Derive $\sum_{r=1}^n r = \frac{n(n+1)}{2}$. (c) Show $\sum_{r=1}^n r(r+1) = \frac{n(n+1)(n+2)}{3}$. [9 marks]
- Q9.** Show $\sum_{r=1}^n r \cdot 2^r = (n-1) \cdot 2^{n+1} + 2$ using the method of differences. [7 marks]
- Q10.** Given $S_n = \frac{3n}{n+2}$, find u_n for $n \geq 2$, write down u_1 , and determine convergence. [6 marks]

Hard

Q11. Prove $\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$ by method of differences and hence evaluate $\sum_{r=n+1}^{2n} r^2$. [9 marks]

Q12. Find A such that $f(r) - f(r+1) = \frac{1}{r(r+1)(r+2)(r+3)}$ where $f(r) = \frac{A}{r(r+1)(r+2)}$. Hence sum the series and find S_∞ . [10 marks]

Q13. Prove $(\sum_{r=1}^n r)^2 = \sum_{r=1}^n r^3$ and verify $\sum_{r=1}^n r^3 - \sum_{r=1}^{n-1} r^3 = n^3$. [8 marks]

Q14. Express $\frac{r^2 + r - 1}{r(r+1)(r+2)}$ in partial fractions. Identify the telescoping structure and find a closed form for the sum. [11 marks]

Q15. Partial sums satisfy $3S_n - 5S_{n-1} = 8 \cdot 3^n$, $S_1 = 12$. Find a_n in closed form and determine convergence. [10 marks]

Exploration (Beyond Syllabus)

Q16. A rational cousin of the Basel problem.

(a) Show $\frac{1}{r^2 - 1} = \frac{1}{2} \left(\frac{1}{r-1} - \frac{1}{r+1} \right)$ for $r \geq 2$. [2]

(b) Write out the telescoping structure for $\sum_{r=2}^n \frac{1}{r^2 - 1}$ explicitly. [3]

(c) Show $\sum_{r=2}^n \frac{1}{r^2 - 1} = \frac{1}{2} \left[1 + \frac{1}{2} - \frac{1}{n} - \frac{1}{n+1} \right]$ and deduce $\sum_{r=2}^\infty \frac{1}{r^2 - 1} = \frac{3}{4}$. [7]

(d) Use a comparison argument to show $\sum_{r=1}^\infty \frac{1}{r^2} < \frac{7}{4}$. Reflect on why the telescoping method cannot yield the Basel sum $\pi^2/6$. [5]

[17 marks]

Q17. Factorial telescoping and an exponential connection.

(a) Compute $E_n = \sum_{k=0}^n \frac{1}{k!}$ for $n = 0, 1, 2, 3, 4$. [2]

(b) Show the tail $T_n = e - E_n \leq \frac{1}{n! \cdot n}$ by bounding via a geometric series. [5]

(c) Using $(k+1)! E_k - k! E_{k-1}$, derive $\sum_{k=1}^n k \cdot k! = (n+1)! - 1$. [7]

(d) Verify for $n = 1, 2, 3$ and hence find $\sum_{r=1}^\infty \frac{k \cdot k!}{(k+2)!}$. [7]

[21 marks]

Chapter 4

Matrices 1

A *matrix* is a rectangular array of numbers. This chapter covers operations, inverses, determinants, and geometric transformations represented by 2×2 and 3×3 matrices.

12 Matrix Operations

12.1 Addition, Scalar Multiplication, and Matrix Multiplication

Result 12.1

For matrices of compatible sizes:

$$(A + B)_{ij} = A_{ij} + B_{ij}, \quad (kA)_{ij} = kA_{ij}, \quad (AB)_{ij} = \sum_k A_{ik}B_{kj}.$$

Matrix multiplication is *associative* and *distributive*, but in general **not commutative**: $AB \neq BA$.

Example 12.1 — Multiply $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$

$$AB = \begin{pmatrix} 1 \cdot 0 + 2 \cdot (-1) & 1 \cdot 1 + 2 \cdot 2 \\ 3 \cdot 0 + 4 \cdot (-1) & 3 \cdot 1 + 4 \cdot 2 \end{pmatrix} = \begin{pmatrix} -2 & 5 \\ -4 & 11 \end{pmatrix}.$$

$$BA = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix} \neq AB. \quad (\text{Non-commutative.})$$

12.2 Transpose and Special Matrices

Result 12.2

The *transpose* A^T is obtained by reflecting across the main diagonal: $(A^T)_{ij} = A_{ji}$.

- *Symmetric*: $A^T = A$.
- *Identity* I : $IA = AI = A$ for all compatible A .
- *Zero matrix* O : all entries zero.

13 The Inverse Matrix

Definition 13.1. The inverse of a square matrix A is the matrix A^{-1} (if it exists) such that $AA^{-1} = A^{-1}A = I$.

Result 13.1 — Inverse of a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad ad - bc \neq 0.$$

Derivation

Let $A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Then:

$$AA^{-1} = \frac{1}{ad - bc} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix} = I. \quad \square$$

Result 13.2 — Properties of inverses

$$(AB)^{-1} = B^{-1}A^{-1}, \quad (A^T)^{-1} = (A^{-1})^T, \quad (A^{-1})^{-1} = A.$$

Example 13.1 — Find A^{-1} for $A = \begin{pmatrix} 3 & 1 \\ 5 & 2 \end{pmatrix}$

$$\det A = 6 - 5 = 1. \text{ So } A^{-1} = \begin{pmatrix} 2 & -1 \\ -5 & 3 \end{pmatrix}. \text{ Check: } AA^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \checkmark$$

14 Determinants

14.1 2×2 Determinant

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

14.2 3×3 Determinant — Cofactor Expansion

Result 14.1

Expanding along the first row:

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg).$$

Geometric meaning

$\det A$ equals the (signed) scale factor by which the matrix A multiplies areas (2×2) or volumes (3×3). If $\det A = 0$, the transformation is *singular* (collapses dimension) and A^{-1} does not exist.

Result 14.2 — Determinant properties

$$\det(AB) = \det A \cdot \det B, \quad \det(A^{-1}) = \frac{1}{\det A}, \quad \det(A^T) = \det A.$$

Example 14.1 — 3×3 determinant

$$\det \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 1 & -1 & 4 \end{pmatrix} = 2(12 - (-2)) - 1(0 - 2) + (-1)(0 - 3) = 2(14) - 1(-2) + (-1)(-3) = 28 + 2 + 3 = 33.$$

15 Matrix Transformations

Common 2×2 transformation matrices (acting on column vectors $\begin{pmatrix} x \\ y \end{pmatrix}$):

Result 15.1

Transformation	Matrix
Rotation by θ anticlockwise about origin	$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$
Reflection in $y = x$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Reflection in x -axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Enlargement scale factor k	$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$
Shear (factor k) parallel to x -axis	$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$

Example 15.1 — Find the image of triangle with vertices $O(0,0)$, $A(1,0)$, $B(0,1)$ under $M = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$

$$MO = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad MA = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad MB = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

$$\text{Area of image} = |\det M| \times \text{area of original} = 6 \times \frac{1}{2} = 3.$$

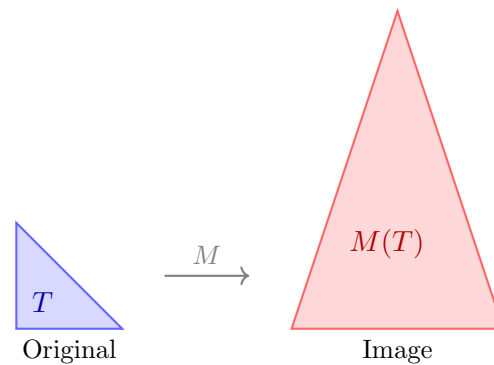


Figure 3. A unit triangle mapped by $M = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$. Area scales by $|\det M| = 6$.

Exercise — Chapter 4 — Matrices 1

Basic

Q1. Given $A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 5 \\ 0 & -2 \end{pmatrix}$, find: (a) $A + B$; (b) AB ; (c) BA ; (d) A^2 . [6 marks]

Q2. Find the inverse of each matrix, or state that it is singular: (a) $\begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix}$; (b) $\begin{pmatrix} 6 & 3 \\ 4 & 2 \end{pmatrix}$; (c) $\begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}$. [6 marks]

Q3. Evaluate: $\det \begin{pmatrix} 3 & -1 \\ 2 & 5 \end{pmatrix}$ and $\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 2 \end{pmatrix}$. [5 marks]

Q4. Write down the 2×2 matrix for: (a) rotation by 90° anticlockwise; (b) reflection in the y -axis; (c) enlargement scale factor 3. [4 marks]

Q5. Use matrix multiplication to solve the system $2x - y = 5$, $x + 3y = -1$ by writing it as $A\mathbf{x} = \mathbf{b}$ and computing $\mathbf{x} = A^{-1}\mathbf{b}$. [5 marks]

Medium

Q6. Show that if $A^2 = A$ (idempotent), then $\det A = 0$ or $\det A = 1$, and give an example of each. [6 marks]

Q7. The transformation T_1 is a rotation by 60° and T_2 is a reflection in the line $y = x \tan 30^\circ$. Find the matrix of $T_2 \circ T_1$ and describe the combined transformation geometrically. [8 marks]

Q8. Prove that for invertible A and B : (a) $(AB)^{-1} = B^{-1}A^{-1}$; (b) $\det(AB) = \det A \cdot \det B$. [8 marks]

Q9. Find all values of k for which $M = \begin{pmatrix} k & 2 & 0 \\ 1 & k & 1 \\ 0 & 3 & k \end{pmatrix}$ is singular. [7 marks]

Q10. A transformation T maps the unit square to a parallelogram of area 10. Given that T has matrix $\begin{pmatrix} a & 2 \\ 1 & a \end{pmatrix}$, find the possible values of a . [5 marks]

Hard

Q11. Prove the Cayley–Hamilton theorem for 2×2 matrices: every matrix satisfies its own characteristic equation $\det(A - \lambda I) = 0$. That is, if $\lambda^2 - (\operatorname{tr} A)\lambda + \det A = 0$ is the characteristic equation, show $A^2 - (\operatorname{tr} A)A + (\det A)I = O$. [8 marks]

Q12. Let $A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$. Prove by induction that $A^n = \begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix}$. [8 marks]

Q13. The matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ represents a transformation that maps the line $y = 2x$ to itself and the line $y = x$ to itself. Find all possible matrices M (up to a scalar multiple). [9 marks]

Q14. Show that if A and B are both $n \times n$ invertible matrices and $AB = BA$, then $(A + B)^{-1}$ exists iff $A + B$ is invertible, and find $(A + B)^{-1}$ in terms of A^{-1} and B^{-1} when $A + B$ is invertible. [8 marks]

Q15. A matrix R satisfies $R^T = R^{-1}$ (orthogonal matrix). Prove: (a) $|\det R| = 1$; (b) the eigenvalues of R have modulus 1; (c) if R is 2×2 real and $\det R = 1$, then R is a rotation matrix. [9 marks]

Exploration (Beyond Syllabus)

Q16. Matrix exponentials and the connection to transformations.

For a square matrix A , define $e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$

(a) Let $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Show $J^2 = -I$, $J^3 = -J$, $J^4 = I$, establishing the period-4 cycle. [3]

(b) Using the cycle, show $e^{\theta J} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ by grouping even and odd terms of the series. [6]

(c) Deduce that $e^{\theta J} \cdot e^{\phi J} = e^{(\theta+\phi)J}$ and interpret geometrically. [3]

[12 marks]

Q17. Determinants and signed areas — a generalisation.

(a) Prove that the (signed) area of a triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ is $\frac{1}{2} \det \begin{pmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{pmatrix}$. [4]

(b) Hence express the area of a polygon with n vertices as a sum of determinants (the shoelace formula). [4]

(c) Prove that three points are collinear iff $\det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} = 0$. [4]

[12 marks]

Chapter 5

Polar Coordinates

Polar coordinates (r, θ) locate a point by its distance from the origin (pole) and angle from the initial line (positive x -axis). They are particularly effective for curves with rotational or radial symmetry.

16 The Polar System

16.1 Conversion Between Polar and Cartesian

Result 16.1

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r = \sqrt{x^2 + y^2}, \quad \theta = \arctan\left(\frac{y}{x}\right).$$

By convention $r \geq 0$ and $\theta \in (-\pi, \pi]$ or $[0, 2\pi)$.

16.2 Common Polar Curves

Result 16.2

Name	Equation	Notes
Circle (centre O)	$r = a$	radius a
Circle through O	$r = 2a \cos \theta$	centre $(a, 0)$
Cardioid	$r = a(1 + \cos \theta)$	heart-shaped
Rose (4 petals)	$r = a \cos 2\theta$	petals at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$
Lemniscate	$r^2 = a^2 \cos 2\theta$	figure-of-eight
Archimedean spiral	$r = a\theta$	

Example 16.1 — Sketch $r = 2(1 + \cos \theta)$

At $\theta = 0$: $r = 4$ (rightmost point). At $\theta = \pi/2$: $r = 2$. At $\theta = \pi$: $r = 0$ (cusp at origin). The curve is a cardioid symmetric about the initial line.

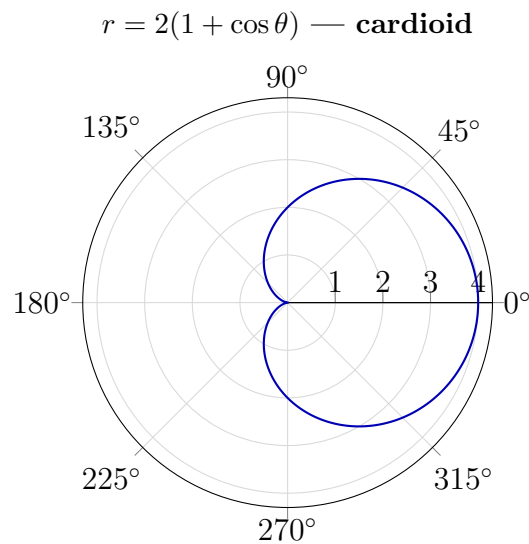


Figure 4. Cardioid $r = 2(1 + \cos \theta)$.

17 Applications of Polar Coordinates

17.1 Area in Polar Coordinates

Result 17.1

The area enclosed by the curve $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$ is

$$A = \frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta.$$

Derivation

A thin sector of angle $d\theta$ and radius r has area $\approx \frac{1}{2}r^2 d\theta$. Integrating gives the result. $\square$

Example 17.1 — Area enclosed by the cardioid $r = a(1 + \cos \theta)$

By symmetry integrate from 0 to π and double:

$$\begin{aligned} A &= 2 \cdot \frac{1}{2} \int_0^{\pi} a^2(1 + \cos \theta)^2 d\theta = a^2 \int_0^{\pi} (1 + 2\cos \theta + \cos^2 \theta) d\theta \\ &= a^2 \int_0^{\pi} \left(1 + 2\cos \theta + \frac{1 + \cos 2\theta}{2} \right) d\theta \\ &= a^2 \left[\theta + 2\sin \theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{\pi} = a^2 \cdot \frac{3\pi}{2} = \frac{3\pi a^2}{2}. \end{aligned}$$

Example 17.2 — Area of one petal of $r = \cos 2\theta$

One petal lies between $\theta = -\pi/4$ and $\theta = \pi/4$:

$$A = \frac{1}{2} \int_{-\pi/4}^{\pi/4} \cos^2 2\theta \, d\theta = \frac{1}{2} \int_{-\pi/4}^{\pi/4} \frac{1 + \cos 4\theta}{2} \, d\theta = \frac{1}{4} \left[\theta + \frac{\sin 4\theta}{4} \right]_{-\pi/4}^{\pi/4} = \frac{\pi}{8}.$$

Exercise — Chapter 5 — Polar Coordinates**Basic**

- Q1.** Convert to Cartesian: (a) $(3, \pi/3)$; (b) $(2, 3\pi/4)$; (c) $(1, -\pi/6)$. [4 marks]
- Q2.** Convert to polar ($r > 0$, $\theta \in (-\pi, \pi]$): (a) $(1, \sqrt{3})$; (b) $(-2, 2)$; (c) $(0, -5)$. [4 marks]
- Q3.** Sketch the curves: (a) $r = 3$; (b) $\theta = \pi/4$; (c) $r = 4 \sin \theta$. Label key points on each. [5 marks]
- Q4.** Show that $r = 2 \cos \theta$ is a circle and find its Cartesian equation, centre, and radius. [4 marks]
- Q5.** Find the area enclosed by $r = 2$ between $\theta = 0$ and $\theta = \pi/2$. Verify by geometry. [4 marks]

Medium

- Q6.** Sketch the cardioid $r = 1 + \sin \theta$, identifying the cusp and maximum r . Find the total enclosed area. [8 marks]
- Q7.** Find the area enclosed by the curve $r^2 = 4 \cos 2\theta$. State the domain of θ for which the curve exists. [7 marks]
- Q8.** Find the points of intersection of $r = 2$ and $r = 3 - 2 \cos \theta$. Hence find the area of the region inside $r = 2$ but outside $r = 3 - 2 \cos \theta$. [9 marks]
- Q9.** Show that the length of the arc of $r = a\theta$ from $\theta = 0$ to $\theta = \Theta$ is $\frac{a}{2} [\Theta \sqrt{1 + \Theta^2} + \ln(\Theta + \sqrt{1 + \Theta^2})]$. [8 marks]
- Q10.** For the lemniscate $r^2 = a^2 \cos 2\theta$, find the total area and the Cartesian equation. [6 marks]

Hard

- Q11.** The curve $r = a(1 + \cos \theta)$ and the circle $r = \frac{3a}{2}$ intersect. Find: (a) their intersection points; (b) the area of the region inside the circle but outside the cardioid;

(c) the area inside the cardioid but outside the circle. [12 marks]

Q12. Prove that the area between the two loops of the limaçon $r = 1 + 2 \cos \theta$ (the region inside the outer loop but outside the inner loop) is $\pi + \frac{3\sqrt{3}}{2}$. [10 marks]

Q13. Show that the curve $r = \sec \theta + 2$ has a vertical asymptote. Find the asymptote and sketch the curve. [8 marks]

Q14. By converting to Cartesian coordinates, show that the spiral $r = e^\theta$ has the property that the angle between the tangent and the radius vector is constant. Find this angle. [9 marks]

Q15. Find the area common to the two circles $r = 2 \sin \theta$ and $r = 2 \cos \theta$, giving your answer in exact form. [8 marks]

Exploration (Beyond Syllabus)

Q16. Arc length in polar form.

The arc length of a polar curve is $L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$.

(a) Derive this formula from the Cartesian arc length formula $L = \int \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta$ using $x = r \cos \theta$, $y = r \sin \theta$. [5]

(b) Use the formula to find the perimeter of the cardioid $r = a(1 + \cos \theta)$. [Answer: $8a$.] [7]

(c) Show that for the Archimedean spiral $r = a\theta$, the arc length from $\theta = 0$ to $\theta = \Theta$ can be expressed as an integral and evaluate it using the substitution $\theta = \sinh t$. [6]

[18 marks]

Q17. Polar curves and complex numbers.

Every point (r, θ) in the plane corresponds to the complex number $z = re^{i\theta} = r(\cos \theta + i \sin \theta)$.

(a) Show that the cardioid $r = a(1 + \cos \theta)$ in polar form corresponds, in complex notation, to the image of the unit circle under the map $w = a \left(z + \frac{1}{z} + 2 \right) / 2$ (where $|z| = 1$). [5]

(b) A four-petalled rose is given by $r = \cos 2\theta$. Write $r^2 = r \cos 2\theta = r \cos^2 \theta - r \sin^2 \theta$ and hence find its Cartesian equation in the form $(x^2 + y^2)^n = f(x, y)$. [4]

(c) For the curve $r^n = a^n \cos n\theta$, show that the total area is $\frac{\pi a^2}{2}$ for all $n \in \mathbb{N}$ with $n \geq 2$. [5]

[14 marks]

Chapter 6

Vectors

This chapter extends vectors to three dimensions, introducing the vector (cross) product, equations of lines, and equations of planes.

18 The Vector Product

Definition 18.1. The vector product (cross product) of $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ is

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}},$$

where θ is the angle between them and $\hat{\mathbf{n}}$ is the unit normal (right-hand rule).

Result 18.1 — Component form

$$\mathbf{a} \times \mathbf{b} = \det \begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}.$$

Result 18.2 — Properties

- Anti-commutative: $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$.
- $\mathbf{a} \times \mathbf{a} = \mathbf{0}$.
- $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin \theta = \text{area of parallelogram spanned by } \mathbf{a}, \mathbf{b}$.
- $\mathbf{a} \times \mathbf{b} = \mathbf{0} \iff \mathbf{a} \parallel \mathbf{b}$.

Example 18.1 — Find $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = (2, 1, -1)$, $\mathbf{b} = (1, 3, 2)$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -1 \\ 1 & 3 & 2 \end{vmatrix} = \mathbf{i}(2 + 3) - \mathbf{j}(4 + 1) + \mathbf{k}(6 - 1) = (5, -5, 5).$$

$$\text{Area of parallelogram} = |(5, -5, 5)| = 5\sqrt{3}.$$

19 Vector Equation of a Line

Result 19.1

The line through point $\mathbf{a}$ with direction $\mathbf{d}$ has vector equation

$$\mathbf{r} = \mathbf{a} + t\mathbf{d}, \quad t \in \mathbb{R}.$$

In component (parametric) form with $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{d} = (d_1, d_2, d_3)$:

$$\frac{x - a_1}{d_1} = \frac{y - a_2}{d_2} = \frac{z - a_3}{d_3} = t \quad (\text{Cartesian/symmetric form}).$$

Example 19.1 — Line through $A(1, 2, -1)$ and $B(3, 0, 4)$

Direction: $\mathbf{d} = \overrightarrow{AB} = (2, -2, 5)$.

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix}, \quad \frac{x-1}{2} = \frac{y-2}{-2} = \frac{z+1}{5}.$$

19.1 Angle Between Two Lines

The acute angle θ between lines with directions $\mathbf{d}_1, \mathbf{d}_2$:

$$\cos \theta = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1||\mathbf{d}_2|}.$$

20 Planes**20.1 Vector and Cartesian Equations****Result 20.1**

The plane through point $\mathbf{a}$ with normal $\mathbf{n} = (n_1, n_2, n_3)$:

$$\mathbf{n} \cdot \mathbf{r} = \mathbf{n} \cdot \mathbf{a} \iff n_1x + n_2y + n_3z = d, \quad d = \mathbf{n} \cdot \mathbf{a}.$$

The plane through three non-collinear points A, B, C :

$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC}.$$

20.2 Distances**Result 20.2**

Distance from point $P(\mathbf{p})$ to plane $\mathbf{n} \cdot \mathbf{r} = d$:

$$\text{dist} = \frac{|\mathbf{n} \cdot \mathbf{p} - d|}{|\mathbf{n}|}.$$

Example 20.1 — Distance from $P(1, 1, 1)$ to $2x - y + 2z = 5$

$$\text{dist} = \frac{|2 - 1 + 2 - 5|}{\sqrt{4 + 1 + 4}} = \frac{|-2|}{3} = \frac{2}{3}.$$

Example 20.2 — Find the equation of the plane through $A(1, 0, 0)$, $B(0, 2, 0)$, $C(0, 0, 3)$

$$\begin{aligned} \overrightarrow{AB} &= (-1, 2, 0), \quad \overrightarrow{AC} = (-1, 0, 3). \\ \mathbf{n} &= \overrightarrow{AB} \times \overrightarrow{AC} = (6, 3, 2). \end{aligned}$$

Plane: $6x + 3y + 2z = 6$ (check: A gives $6 \cdot 1 = 6$. ✓)

Exercise — Chapter 6 — Vectors

Basic

Q1. Find $\mathbf{a} \times \mathbf{b}$ for: (a) $\mathbf{a} = (1, 2, 3)$, $\mathbf{b} = (4, 5, 6)$; (b) $\mathbf{a} = (2, -1, 1)$, $\mathbf{b} = (1, 3, -2)$. Verify $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0$ in each case. [6 marks]

Q2. Find the vector equation and Cartesian form of the line through $(2, -1, 3)$ with direction $(1, 4, -2)$. [4 marks]

Q3. Find the equation of the plane with normal $(3, 1, -2)$ passing through the point $(1, 0, 4)$. [3 marks]

Q4. Find the distance from the point $(2, 1, -3)$ to the plane $x - 2y + 2z = 6$. [4 marks]

Q5. Determine whether the lines $\mathbf{r} = (1, 0, 2) + t(2, 1, -1)$ and $\mathbf{r} = (3, 2, 0) + s(1, -1, 1)$ intersect. If so, find the intersection point. [5 marks]

Medium

Q6. Find the equation of the plane through $A(1, 2, 3)$, $B(3, -1, 2)$, $C(2, 1, -1)$. Find the distance from the origin to this plane. [8 marks]

Q7. Two planes are given by $\Pi_1 : 2x - y + z = 3$ and $\Pi_2 : x + y - z = 1$. Find: (a) the angle between them; (b) the vector equation of their line of intersection. [9 marks]

Q8. Show that the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x+1}{1} = \frac{y-1}{-1} = \frac{z}{2}$ are skew. Find the shortest distance between them. [10 marks]

Q9. The points $A(2, 0, 1)$, $B(0, 3, -1)$, $C(1, 1, 2)$, D form a parallelogram. Find D and the area of the parallelogram. [7 marks]

Q10. A line ℓ has equation $\mathbf{r} = \mathbf{a} + t\mathbf{d}$. Show that the foot of the perpendicular from point P to ℓ is at parameter $t^* = \frac{(\mathbf{p} - \mathbf{a}) \cdot \mathbf{d}}{|\mathbf{d}|^2}$. [5 marks]

Hard

Q11. Prove the vector triple product identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$ by expanding in components. [8 marks]

Q12. Using the scalar triple product $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$, prove that four points A, B, C, D are coplanar iff $[\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}] = 0$. Test whether $(1, 1, 0)$, $(2, 0, 1)$, $(0, 2, 1)$, $(3, 1, 2)$ are coplanar. [9 marks]

Q13. Prove the formula for the distance between two skew lines $\mathbf{r} = \mathbf{a} + t\mathbf{d}_1$ and $\mathbf{r} = \mathbf{b} + s\mathbf{d}_2$ is $d = \frac{|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|}{|\mathbf{d}_1 \times \mathbf{d}_2|}$. [8 marks]

Q14. The plane Π passes through the line of intersection of $x + 2y - z = 3$ and $2x - y + z = 1$, and also through the point $(1, 1, 1)$. Find the equation of Π . [7 marks]

Q15. Prove that the angle between a line with direction $\mathbf{d}$ and a plane with normal $\mathbf{n}$ satisfies $\sin \psi = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$. Hence find the angle between the line $\frac{x-1}{2} = \frac{y}{1} = \frac{z+1}{-2}$ and the plane $x + y + z = 0$. [8 marks]

Exploration (Beyond Syllabus)

Q16. Plucker coordinates and lines in 3D.

Any line in 3D can be encoded by a pair of vectors: its direction $\mathbf{d}$ and its moment $\mathbf{m} = \mathbf{a} \times \mathbf{d}$ (where $\mathbf{a}$ is any point on the line).

- (a) Show that $\mathbf{m} \cdot \mathbf{d} = 0$ always, and that $\mathbf{m}$ is independent of the choice of $\mathbf{a}$. [4]
- (b) Show that two lines $(\mathbf{d}_1, \mathbf{m}_1)$ and $(\mathbf{d}_2, \mathbf{m}_2)$ intersect iff $\mathbf{d}_1 \cdot \mathbf{m}_2 + \mathbf{d}_2 \cdot \mathbf{m}_1 = 0$. [5]
- (c) Rederive the skew-distance formula in terms of Plucker coordinates. [4]

[13 marks]

Q17. Geometric interpretation of the scalar triple product.

- (a) Prove that $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$ equals the signed volume of the parallelepiped with edge vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$. [5]
- (b) Using this, show that the volume of a tetrahedron $ABCD$ is $\frac{1}{6} |[\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}]|$. [3]
- (c) A regular tetrahedron has edge length a . Find its volume using vectors (without tables). [5]

[13 marks]

Chapter 7

Proof by Induction

Mathematical induction is a method of proof for statements involving a natural number parameter n . It is logically equivalent to the well-ordering principle of $\mathbb{N}$.

21 The Inductive Process

Result 21.1 — Structure of a proof by induction

To prove $P(n)$ for all $n \geq n_0$:

1. **Base case:** verify $P(n_0)$ directly.
2. **Inductive step:** assume $P(k)$ holds for some $k \geq n_0$ (inductive hypothesis), then *prove* $P(k+1)$.
3. **Conclusion:** by the principle of mathematical induction, $P(n)$ holds for all $n \geq n_0$.

21.1 Summation Formulae

Example 21.1 — Prove $\sum_{k=1}^n k = \frac{n(n+1)}{2}$ by induction

Base case ($n = 1$): LHS = 1, RHS = $\frac{1 \cdot 2}{2} = 1$. ✓

Inductive step: Assume $\sum_{k=1}^m k = \frac{m(m+1)}{2}$ for some $m \geq 1$.

Then

$$\begin{aligned} \sum_{k=1}^{m+1} k &= \frac{m(m+1)}{2} + (m+1) \\ &= (m+1) \left(\frac{m}{2} + 1 \right) \\ &= \frac{(m+1)(m+2)}{2}, \end{aligned}$$

which is the formula with $n = m+1$. □

Example 21.2 — Prove $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ by induction

Base case ($n = 1$): LHS = 1, RHS = $\frac{1 \cdot 2 \cdot 3}{6} = 1$. ✓

Inductive step: Assume truth for m . Then

$$\begin{aligned}
 \sum_{k=1}^{m+1} k^2 &= \frac{m(m+1)(2m+1)}{6} + (m+1)^2 \\
 &= \frac{(m+1)}{6} [m(2m+1) + 6(m+1)] \\
 &= \frac{(m+1)(2m^2 + 7m + 6)}{6} \\
 &= \frac{(m+1)(m+2)(2m+3)}{6},
 \end{aligned}$$

which is the formula with $n = m + 1$. $\square$

21.2 Matrix Powers

Example 21.3 — Prove $A^n = \begin{pmatrix} 1 & 2n \\ 0 & 1 \end{pmatrix}$ **for** $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

Base case ($n = 1$): $A^1 = A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. $\checkmark$

Inductive step: Assume $A^m = \begin{pmatrix} 1 & 2m \\ 0 & 1 \end{pmatrix}$. Then

$$A^{m+1} = A^m \cdot A = \begin{pmatrix} 1 & 2m \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2m+2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2(m+1) \\ 0 & 1 \end{pmatrix}. \square$$

22 Proof by Induction for Divisibility

Example 22.1 — Prove $f(n) = 5^n - 1$ **is divisible by 4 for all** $n \geq 1$

Base case: $f(1) = 4$. Divisible by 4. $\checkmark$

Inductive step: Assume $4 \mid 5^m - 1$, i.e. $5^m = 4q + 1$ for some integer q . Then

$$5^{m+1} - 1 = 5 \cdot 5^m - 1 = 5(4q + 1) - 1 = 20q + 4 = 4(5q + 1).$$

Hence $4 \mid 5^{m+1} - 1$. $\square$

Example 22.2 — Prove $n^3 + 2n$ **is divisible by 3**

Base case: $n = 1$: $1 + 2 = 3$. $\checkmark$

Inductive step: Assume $3 \mid m^3 + 2m$. Then

$$\begin{aligned}
 (m+1)^3 + 2(m+1) &= m^3 + 3m^2 + 3m + 1 + 2m + 2 \\
 &= (m^3 + 2m) + 3m^2 + 3m + 3 \\
 &= (m^3 + 2m) + 3(m^2 + m + 1).
 \end{aligned}$$

Both terms are divisible by 3 by hypothesis and directly. $\square$

Exercise — Chapter 7 — Proof by Induction

Basic

- Q1.** Prove by induction: $\sum_{k=1}^n r^3 = \left[\frac{n(n+1)}{2} \right]^2$. [6 marks]
- Q2.** Prove by induction: $\sum_{k=1}^n (2r-1) = n^2$. [5 marks]
- Q3.** Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$. [5 marks]
- Q4.** Prove that $4^n + 6n - 1$ is divisible by 9 for all $n \geq 1$. [5 marks]
- Q5.** Given $A = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$, conjecture a formula for A^n and prove it by induction. [5 marks]

Medium

- Q6.** Prove by induction: $\sum_{k=1}^n r(r+1) = \frac{n(n+1)(n+2)}{3}$. [6 marks]
- Q7.** Prove that for all $n \geq 1$: $\prod_{r=2}^n \left(1 - \frac{1}{r^2} \right) = \frac{n+1}{2n}$. [7 marks]
- Q8.** Prove by induction: $2^n > n^2$ for all $n \geq 5$. [7 marks]
- Q9.** Prove by induction: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ for all $n \geq 1$. Hence state $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{100}$. [6 marks]
- Q10.** Prove that $3^{2n+1} + 2^{n+2}$ is divisible by 7 for all $n \geq 0$. [7 marks]

Hard

- Q11.** Prove by strong induction that every integer $n \geq 2$ can be written as a product of primes. [7 marks]
- Q12.** The Fibonacci sequence is defined by $F_1 = F_2 = 1$, $F_{n+2} = F_{n+1} + F_n$.
 (a) Prove $F_1 + F_2 + \cdots + F_n = F_{n+2} - 1$. [5]
 (b) Prove $F_n^2 + F_{n+1}^2 = F_{2n+1}$. [6]
 [11 marks]
- Q13.** Prove by induction that the number of subsets of a set with n elements is 2^n . [7 marks]
- Q14.** Let $u_1 = 1$, $u_{n+1} = \frac{u_n}{2u_n + 1}$. Prove by induction that $u_n = \frac{1}{2n-1}$. [7 marks]

Q15. Prove by induction: $\det \begin{pmatrix} 2 & 1 & 0 & \cdots & 0 \\ 1 & 2 & 1 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & \cdots & 1 & 2 & 1 \\ 0 & \cdots & 0 & 1 & 2 \end{pmatrix}_{n \times n} = n + 1.$ [10 marks]

Exploration (Beyond Syllabus)

Q16. Well-ordering and the equivalence of induction principles.

- (a) State the *well-ordering principle*: every non-empty subset of $\mathbb{N}$ has a least element. Prove that this implies the principle of mathematical induction (without using induction). [6]
- (b) Prove the equivalence: *strong induction* (if $P(1), \dots, P(k)$ imply $P(k+1)$) is equivalent to ordinary induction. [5]
- (c) Use the well-ordering principle directly to prove the Fundamental Theorem of Arithmetic: every integer $n \geq 2$ has a unique prime factorisation (existence via part (a); uniqueness as a separate argument). [7]

[18 marks]

Q17. Induction in algebra — Bernoulli's inequality and beyond.

- (a) Prove Bernoulli's inequality: $(1+x)^n \geq 1+nx$ for all $x \geq -1$ and $n \in \mathbb{N}$. [5]
- (b) Use (a) to show that $\left(1 + \frac{1}{n}\right)^n$ is strictly increasing in n . [4]
- (c) Prove by induction that $\left(1 + \frac{1}{n}\right)^n < 3$ for all $n \geq 1$. [5]
- (d) Hence conclude $e \leq 3$ (where $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$), and state why equality is excluded. [3]

[17 marks]

Key Results — Further Pure Mathematics 1

Topic	Result
Quadratic roots	$\alpha + \beta = -b/a, \alpha\beta = c/a$
Cubic roots	$\sum \alpha = -b/a, \sum \alpha\beta = c/a, \alpha\beta\gamma = -d/a$
$\sum k$	$n(n+1)/2$
$\sum k^2$	$n(n+1)(2n+1)/6$
$\sum k^3$	$[n(n+1)/2]^2$
Telescoping	$\sum [f(r) - f(r-1)] = f(n) - f(0)$
Geometric S_∞	$a/(1-r), r < 1$
2×2 inverse	$(1/\det A) \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$
$\det(AB)$	$\det A \cdot \det B$
Rotation matrix	$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$
Polar area	$\frac{1}{2} \int_\alpha^\beta r^2 d\theta$
Cross product	$ \mathbf{a} \times \mathbf{b} = \mathbf{a} \mathbf{b} \sin \theta$
Point-plane dist	$ \mathbf{n} \cdot \mathbf{p} - d / \mathbf{n} $
Induction base	Always verify $P(n_0)$ explicitly
Divisibility	Write $f(k+1) = f(k) + (\text{multiple of } m)$